UNIT III
IoT Data Link Layer and Network Layer Protocols

Syllabus: PHY/MAC Layer (3GPP MTC, IEEE 802.11, IEEE 802.15), Wireless HART, Z Wave, Bluetooth Low Energy, Zigbee Smart Energy, DASH7 - Network Layer-IPv4, IPv6, 6LoWPAN, 6TiSCH, ND, DHCP, ICMP, RPL, CORPL, CARP PHY/MAC Layer:

3.1. Introduction to PHY/MAC Layer:

In IoT (Internet of Things) architecture, the PHY (Physical) and MAC (Media Access Control) layers refer to the lower layers of the communication stack responsible for establishing and managing the physical connectivity and data transmission between IoT devices.

PHY Layer: The PHY layer is the lowest layer in the IoT communication stack and deals with the physical transmission of data over the communication medium. It defines the hardware and electrical specifications required for transmitting and receiving signals. The PHY layer is responsible for aspects such as modulation, coding, frequency bands, transmission power, and signal propagation. It ensures reliable and efficient transmission of data between IoT devices.

Depending on the specific IoT deployment, various wireless technologies can be used at the PHY layer, including Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRa, Sigfox, Cellular (2G/3G/4G/5G), and others. Each technology has its own PHY layer specifications, such as frequency ranges, transmission power levels, and modulation techniques.

MAC Layer: The MAC layer resides above the PHY layer and is responsible for managing access to the shared communication medium and handling data packet transmission between IoT devices. It defines protocols and rules for devices to access the medium, avoiding collisions and ensuring fair and efficient utilization of the available bandwidth.

The MAC layer controls the timing, synchronization, and channel access methods for IoT devices. It can use various techniques such as contention-based access (e.g., CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance), time-based access (e.g., TDMA - Time Division Multiple Access), or scheduled access (e.g., FDMA - Frequency Division Multiple Access) depending on the specific technology being used.

Additionally, the MAC layer may handle functions such as packet fragmentation, error detection, acknowledgments, and retransmissions to ensure reliable data transmission in the presence of interference or other communication challenges.

Both the PHY and MAC layers play a crucial role in establishing robust and efficient communication between IoT devices. They define the physical and access control mechanisms necessary for transmitting data over various wireless technologies, ensuring reliable and optimized connectivity in IoT deployments.

3.2. 3GPP MTC:

3GPP MTC (Machine Type Communication) in IoT (Internet of Things) refers to the cellular communication technologies and standards developed by the 3rd Generation Partnership Project (3GPP) specifically for IoT devices. It encompasses the specifications and enhancements made to cellular networks to support the unique requirements of IoT applications.

Here are some key aspects of 3GPP MTC in IoT:

Low-Power and Low-Complexity Devices: 3GPP MTC focuses on enabling communication for low-power and low-complexity IoT devices. These devices typically have limited processing capabilities, memory, and power supply, requiring communication technologies that are optimized for efficiency and resource conservation.

Coverage Enhancement: 3GPP MTC technologies, such as LTE-M (LTE for MTC), eMTC (enhanced Machine Type Communication), and NB-IoT (Narrowband IoT), offer coverage enhancements to ensure reliable connectivity in challenging environments. These technologies provide extended coverage range, improved signal penetration through buildings and underground areas, and better performance in remote or rural areas.

Power Efficiency: Power consumption is a critical concern for IoT devices, many of which operate on battery power. 3GPP MTC introduces power-saving features such as extended discontinuous reception (eDRX) and power saving mode (PSM) to minimize energy consumption. These mechanisms allow IoT devices to enter sleep modes for extended periods while still maintaining network connectivity.

Optimized Data Rates: 3GPP MTC technologies provide optimized data rates suitable for IoT applications. While they may not offer the high data throughput of traditional cellular networks, they provide sufficient bandwidth for transmitting small amounts of data, periodic sensor readings, and control messages required by IoT devices.

Quality of Service (QoS): 3GPP MTC supports differentiated QoS for different types of IoT applications. It provides mechanisms to prioritize critical data traffic and ensure reliable transmission for mission-critical applications. QoS parameters can be adjusted to meet the specific requirements of diverse IoT use cases.

Security and Authentication: 3GPP MTC includes robust security mechanisms to protect IoT device communications. It utilizes encryption, authentication, and access control measures to ensure the confidentiality and integrity of data transmitted over the cellular network.

Integration with IoT Platforms: 3GPP MTC technologies are designed to seamlessly integrate with IoT platforms and cloud services. They provide standardized protocols and interfaces for data exchange, device management, and integration with higher-level IoT systems.

By leveraging 3GPP MTC, IoT applications can benefit from the wide coverage, reliable connectivity, power efficiency, and security features offered by cellular networks. These technologies enable cellular communication for a wide range of IoT devices and applications, facilitating the scalability and interoperability of IoT deployments.

Machine-type communication (MTC) is a form of data communication which involves one or more entities that do not necessarily need human interaction.

For MTC communication the following communication scenarios can be identified:

1) Communication scenario with MTC devices communicating with MTC server. MTC server is located in the operator domain

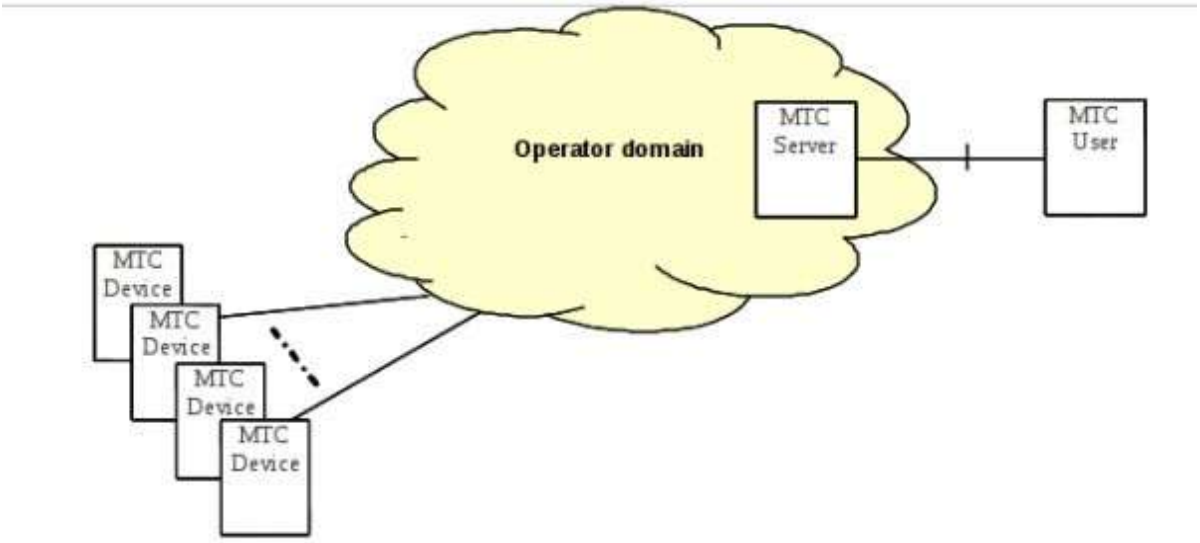


Fig.3.1. 3GPP MTC Architecture – Indirect Model (MTC Server Controlled by the Network Operator / Inside Operator Domain)

2) Communication scenario with MTC devices communicating with MTC server. MTC server is located outside the operator domain:

   The network operator provides network connectivity to MTC Server(s). This applies to MTC Servers controlled by the network operator or to MTC Servers not controlled by the network operator.

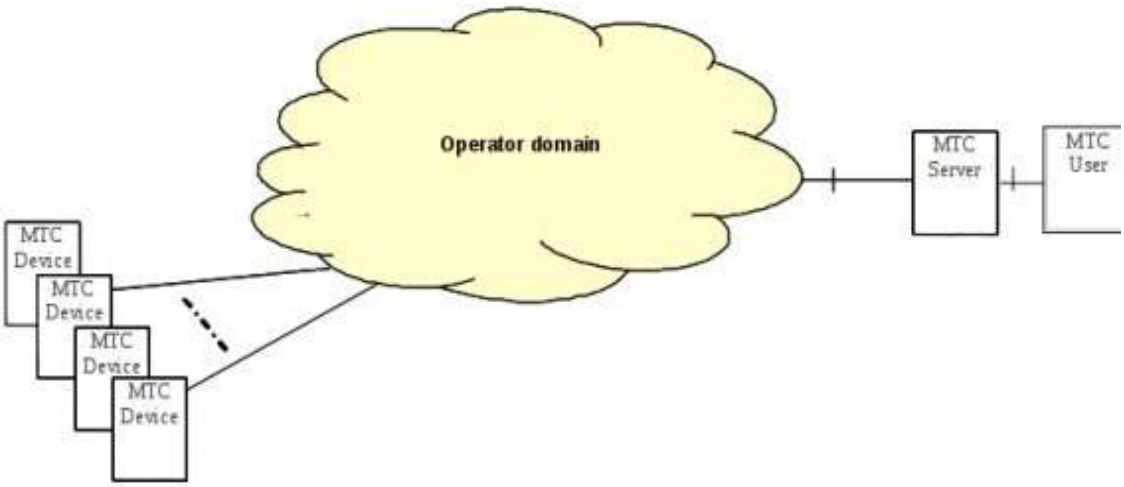


Fig.3.2: 3GPP Machine Type Communication (MTC) Architecture (Indirect Model / MTC Server controlled by MTC User setup).

3) MTC devices communicating with each other:
   The communication scenario where the MTC Devices communicate directly without intermediate MTC Server

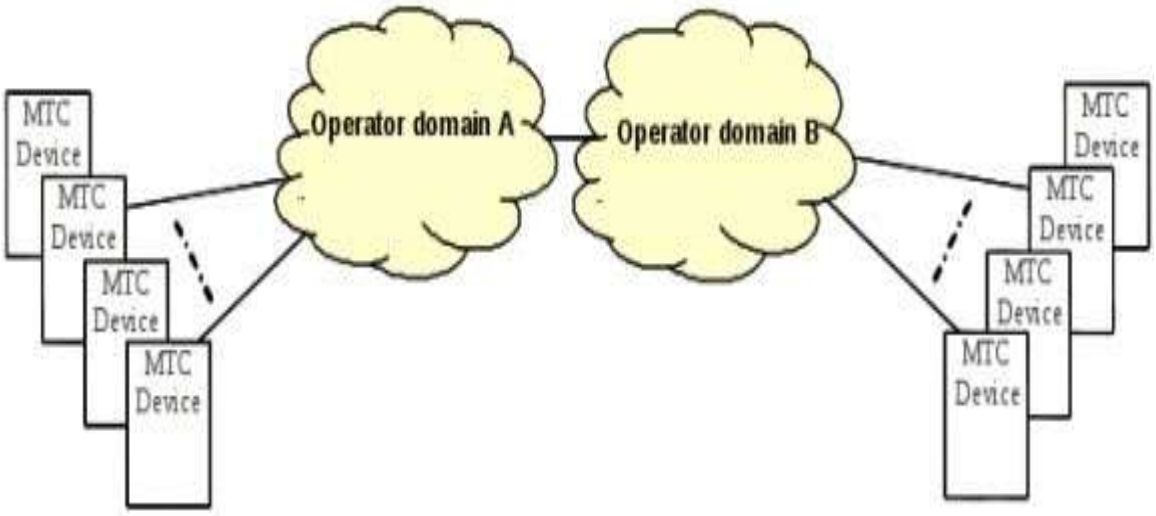


Fig.3.3: 3GPP MTC Architecture – Direct Model (Device-to-Device / Peer-to-Peer MTC).

As technology evolves, there are important changes in capabilities and costs. More computing power, memory and communication capabilities make it possible for machines to take over tasks presently done by, but not well suited to human beings. Lower costs make it practical for machines to take over tasks not well suited to expensive human beings. Increasing capabilities and lower costs together open new opportunities for revenue generating services not previously economical to do.

The increasing capability of machines makes it possible to avoid dull and repetitious work having to be done by people, freeing them to utilize their capabilities and intelligence in better suited and much more fruitful activities.

MTC Device: A MTC Device is a UE equipped for Machine Type Communication, which communicates through a PLMN with MTC Server(s) and/or other MTC Device(s).

MTC Feature: MTC Features are network functions to optimise the network for use by M2M applications.

MTC Group: A MTC Group is a group of MTC Devices that share one or more MTC Features and that belong to the same MTC Subscriber.

MTC Server: A MTC Server is a server, which communicates to the PLMN itself, and to MTC Devices through the PLMN. The MTC Server can also have an interface which can be accessed by the MTC User. The MTC Server can:

Provides services for other servers (e.g. The MTC Server is a Services Capability Server for an Application Server), and/or

Provides services for applications and can host the application (e.g. The MTC Server is an Application Server).

MTC User: A MTC User uses the service provided by the MTC Server.

MTC Subscriber: A MTC Subscriber is a legal entity having a contractual relationship with the network operator to provide service to one or more MTC Devices.

3.3. IEEE 802.11 STANDARDS (WLAN) :

IEEE 802.11 standard, popularly known as WiFi, lays down the architecture and specifications of wireless LANs (WLANs). WiFi or WLAN uses high frequency radio waves for connecting the nodes.

3.3.1. Classification of IEEE 802.11

There are several standards of IEEE 802.11 WLANs. The prominent among them are 802.11, 802.11a, 802.11b, 802.11g, 802.11n and 802.11p. All the standards use carrier-sense multiple access with collision avoidance (CSMA/CA). Also, they have support for both centralized base station based as well as ad hoc networks.

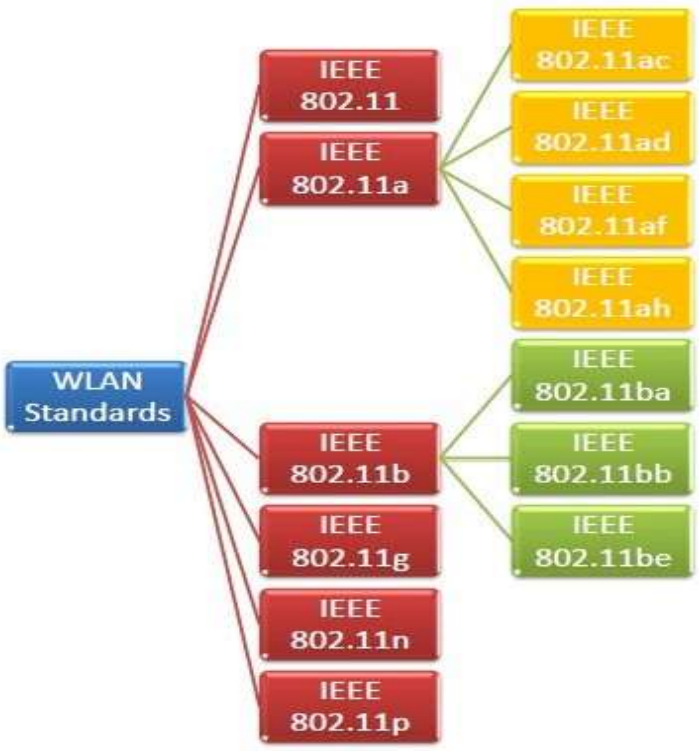


Fig.3.4: Classification of IEEE 802.11 WLAN Standards

IEEE 802.11

IEEE 802.11 was the original version released in 1997. It provided 1 Mbps or 2 Mbps data rate in the 2.4 GHz band and used either frequency-hopping spread spectrum (FHSS) or direct-sequence spread spectrum (DSSS). It is obsolete now.

IEEE 802.11a

802.11a was published in 1999 as a modification to 802.11, with orthogonal frequency division multiplexing (OFDM) based air interface in physical layer instead of FHSS or DSSS of 802.11. It provides a maximum data rate of 54 Mbps operating in the 5 GHz band. Besides it provides error correcting code. As 2.4 GHz band is crowded, relatively sparsely used 5 GHz imparts additional advantage to 802.11a.

Further amendments to 802.11a are 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj etc.

IEEE 802.11b

802.11b is a direct extension of the original 802.11 standard that appeared in early 2000. It uses the same modulation technique as 802.11, i.e. DSSS and operates in the 2.4 GHz band. It has a higher data rate of 11 Mbps as compared to 2 Mbps of 802.11, due to which it was rapidly adopted in wireless LANs. However, since 2.4 GHz band is pretty crowded, 802.11b devices faces interference from other devices.

Further amendments to 802.11b are 802.11ba, 802.11bb, 802.11bc, 802.11bd and 802.11be.

IEEE 802.11g

802.11g was indorsed in 2003. It operates in the 2.4 GHz band (as in 802.11b) and provides a average throughput of 22 Mbps. It uses OFDM technique (as in 802.11a). It is fully backward compatible with 802.11b. 802.11g devices also faces interference from other devices operating in 2.4 GHz band.

IEEE 802.11n

802.11n was approved and published in 2009 that operates on both the 2.4 GHz and the 5 GHz bands. It has variable data rate ranging from 54 Mbps to 600 Mbps. It provides a marked improvement over previous standards 802.11 by incorporating multiple-input multiple-output antennas (MIMO antennas).

IEEE 802.11p

802.11 is an amendment for including wireless access in vehicular environments (WAVE) to support Intelligent Transportation Systems (ITS). They include network communications between vehicles moving at high speed and the environment. They have a data rate of 27 Mbps and operate in 5.9 GHz band.

3.3.2. IEEE 802.11 protocol architecture:

Wireless LANs are those Local Area Networks that use high frequency radio waves instead of cables for connecting the devices in LAN. Users connected by WLANs can move around within the area of network coverage. Most WLANs are based upon the standard IEEE 802.11 or WiFi.

The components of IEEE 802.11 architecture are as follows

1) Stations (STA) − Stations comprise all devices and equipments that are connected to the wireless LAN. A station can be of two types:

Wireless Access Points (WAP)−WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.

Client− Clients are workstations, computers, laptops, printers, smart phones, etc. Each station has a wireless network interface controller.

2) Basic Service Set (BSS) −A basic service set is a group of stations communicating at physical layer level. BSS can be of two categories depending upon mode of operation:

Infrastructure BSS − Here, the devices communicate with other devices through access points.

Independent BSS − Here, the devices communicate in peer-to-peer basis in an ad hoc manner.

3) Extended Service Set (ESS) − It is a set of all connected BSS.

4) Distribution System (DS) − It connects access points in ESS**.**

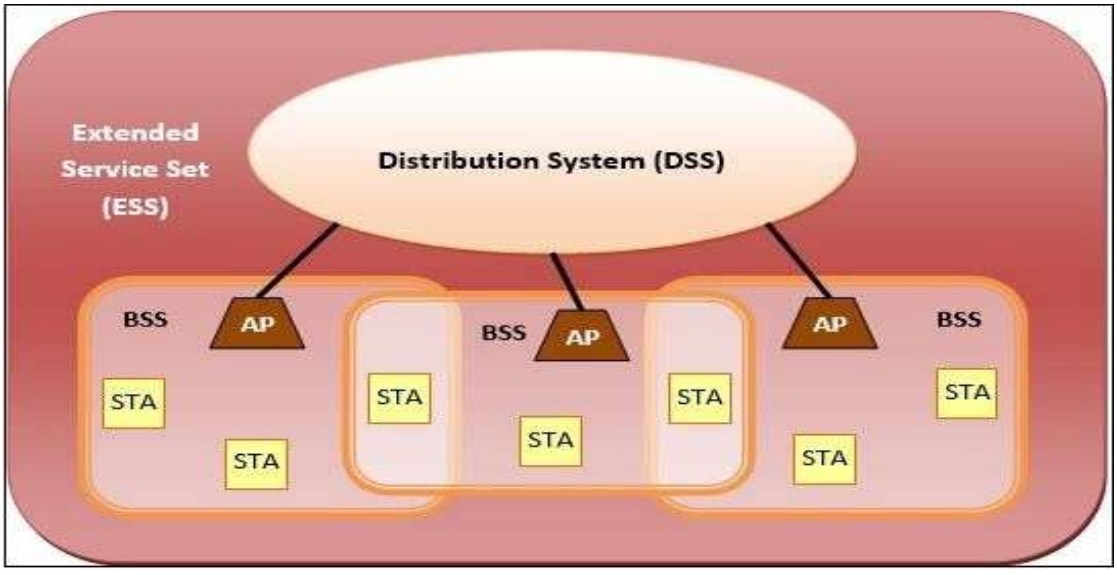


Fig.3.5: Architecture of IEEE 802.11

An ESS can also provide gateway access for wireless users into a wired network. Each end station associates itself with one access point. Above Figure shows three BSSs interconnected through three APs to a distribution system. If station A associated with AP-1 wants to send a frame to another station associated with AP- 2, the first sends a frame to its access point (AP-1), which forwards the frame across the distribution system to the access point AP-2. AP-2 finally delivers it to the destination station.
The technique used for this purpose is known as scanning, which involves the following steps:

- A station sends a probe frame.
- All APs within reach reply with a probe response frame.
- The station selects one of the access points, and sends the AP an Association Request frame.
- The AP replies with an Association Response frame.

3.3.3. Medium Access Control:

Most wired LANs products use Carrier Sense Multiple Access with Collision Detection (CSMA/CD) as the MAC protocol. Carrier Sense means that the station will listen before it transmits. If there is already someone transmitting, then the station waits and tries again later. If no one is transmitting then the station goes ahead and sends what it has. But when more than one station tries to transmit, the transmissions will collide and the information will be lost.

This is where Collision Detection comes into play. The station will listen to ensure that its transmission made it to the destination without collisions. If a collision occurred then the stations wait and try again later. The time the station waits is determined by the back off algorithm. This technique works great for wired LANs but wireless topologies can create a problem for CSMA/CD. However, the wireless medium presents some unique challenges not present in wired LANs that must be dealt with by the MAC used for IEEE
802.11. Some of the challenges are:

- The wireless LAN is prone to more interference and is less reliable.
- The wireless LAN is susceptible to unwanted interception leading to security problems.
- There are so called hidden station and exposed station problems.

In the discussion of both the problem, we shall assume that all radio transmitters have fixed range. When the receiver is in the range of two active transmitters then the signal will be garbled. It is important to note that not all stations are in range of two transmitters.

3.3.4. The Hidden Station Problem

Consider a situation when A is transmitting to B, as depicted in the Fig. If C senses the media, it will not hear anything because it is out of range, and thus will falsely conclude that no transmission is going on and will start transmit to
B. the transmission will interfere at B, wiping out the frame from A.

The problem of a station not been able to detect a potential competitor for the medium because the competitor is too far away is referred as Hidden Station Problem. As in the described scenario C act as a hidden station to A, this is also competing for the medium.

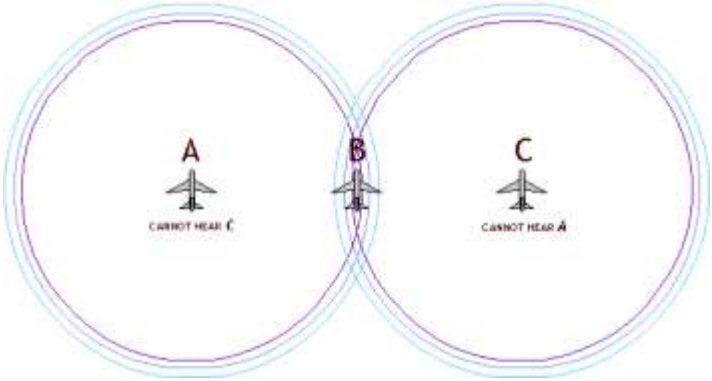


Fig.3.6: Hidden Station Problem in IEEE 802.11

3.3.5. Exposed Station problem

Now consider a different situation where B is transmitting to A, and C sense the medium and detects the ongoing transmission between B and A. C falsely conclude that it cannot transmit to D, when the fact is that such transmission would cause on problem. A transmission could cause a problem only when the destination is in zone between B and C. This problem is referred as Exposed station Problem. In this scenario as B is exposed to C, that's why C assumes it cannot transmit to D. So this problem is known as Exposed station problem (i.e. problem caused due to exposing of a station). The problem here is that before transmission, a station really wants to know that whether or not there is any activity around the receiver. CSMA merely tells whether or not there is any activity around the station sensing the carrier.

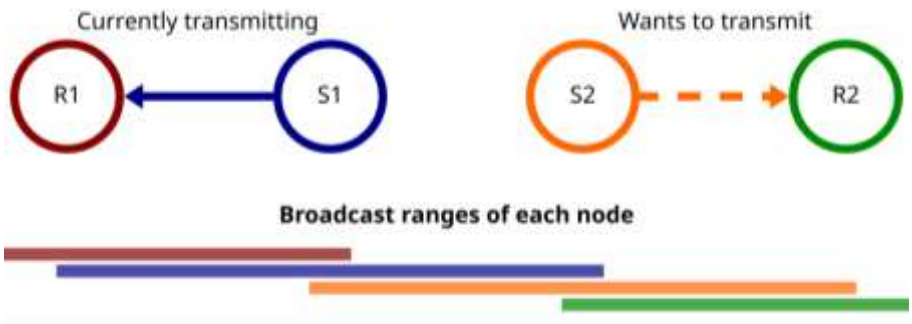


Fig.3.7. Exposed Node Problem in IEEE 802.11

3.3.6. Security

Wireless LANs are subjected to possible breaches from unwanted monitoring. To overcome this problem, IEEE 802.11 specifies an optional MAC layer security system known as Wired Equivalent Privacy (WEP). The objective is to provide a level of privacy to the wireless LAN similar to that enjoyed by wired Ethernets. It is achieved with the help of a 40-bit shared key authentication service. By default each BSS supports up to four 40-bit keys that are shared by all the clients in the BSS. Keys unique to a pair of communicating clients and direction of transmission may also be used. Advanced Encryption Standard (AES) (802.11i) for authentication and encryption is recommended as a long-term solution.

3.3.7. Frame Format of IEEE 802.11

The main fields of a frame of wireless LANs as laid down by IEEE 802.11 are −

Frame Control − It is a 2 bytes starting field composed of 11 subfields. It contains control information of the frame.

Duration − It is a 2-byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.

Address fields − There are three 6-byte address fields containing addresses of source, immediate destination, and final endpoint respectively.

Sequence − It a 2 bytes field that stores the frame numbers.

Data − This is a variable-sized field that carries the data from the upper layers. The maximum size of the data field is 2312 bytes.

Check Sequence − It is a 4-byte field containing error detection information.

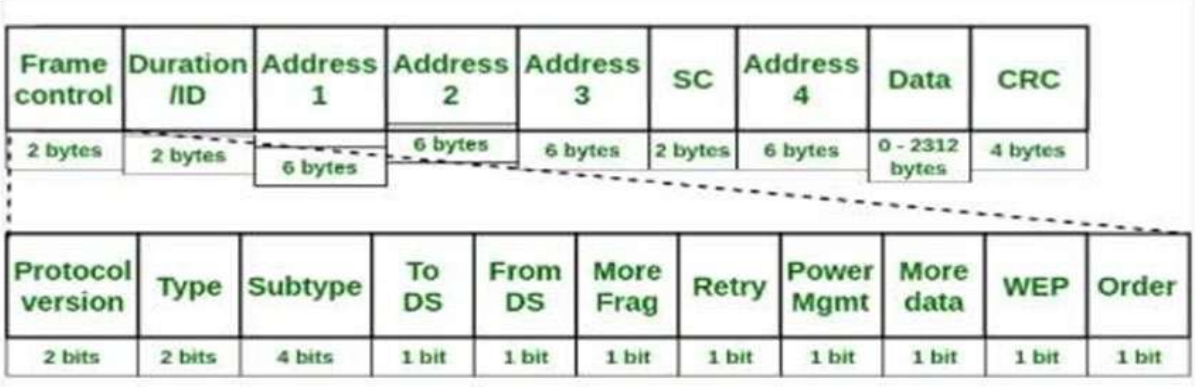


Fig.3.8. IEEE 802.11 Frame Structure

Frame Control(FC) –

It is 2 bytes long field which defines type of frame and some control information. Various fields presentin FC are:

1. **Version:**

It is a 2 bit long field which indicates the current protocol version which is fixed to be 0 for now.

2. Type:

It is a 2 bit long field which determines the function of frame i.e management (00), control (01) ordata (10). The value 11 is reserved.

3. Subtype:

It is a 4 bit long field which indicates sub-type of the frame like 0000 for association request, 1000 forbeacon.

4. To DS:

It is a 1 bit long field which when set indicates that destination frame is for DS(distribution system).

5. From DS:

It is a 1 bit long field which when set indicates frame coming from DS.

6. More frag (More fragments):

It is 1 bit long field which when set to 1 means frame is followed by other fragments.

7. Retry:

It is 1-bit long field, if the current frame is a retransmission of an earlier frame, this bit is set to 1.

8. Power Mgmt (Power management):

It is 1-bit long field that indicates the mode of a station after successful transmission of a frame. Set to 1 the field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.

9. More data:

It is 1-bit long field that is used to indicate receiver that a sender has more data. to send than the current frame. This can be used by an access point to indicate to a station in power-save mode that more packets are buffered or it can be used by a station to indicate to an access point after being polled that more polling is necessary as the station has more data ready to transmit.

10. WEP:

It is 1 bit long field which indicates that the standard security mechanism of 802.11 is applied.

11. Order:

It is 1 bit long field, if this bit is set to 1 the received frames must be processed in strict order.

3.4. IEEE 802.15 Standards (WPAN/ WSN):

IEEE 802.15 is a standard for wireless personal area networks (WPANs) or Wireless

Specialty Networks(WSNs) developed by the Institute of Electrical and Electronics Engineers (IEEE). It defines the physical layer (PHY) and medium access control (MAC) layer specifications for short-range wireless communication.

3.4.1. IEEE 802.15 Family

The IEEE 802.15 architecture consists of multiple task groups, each focusing on specific applications and requirements. Here are some key task groups within the IEEE 802.15 standard:

IEEE 802.15.1 (Bluetooth): This task group defines the specifications for Bluetooth wireless technology, which enables short-range communication between devices such as mobile phones, laptops, and peripherals.

IEEE 802.15.3: This task group focuses on high-rate wireless personal area networks (HR-WPANs). It defines the PHY and MAC layers for applications that require higher data rates, such as streaming multimedia.

IEEE 802.15.4: This task group specifies the PHY and MAC layers for low-rate wireless personal area networks (LR-WPANs). It is commonly used in applications like home automation, industrial control, and wireless sensor networks. The most well-known standard built on IEEE 802.15.4 is Zigbee.

IEEE 802.15.6: This task group concentrates on wireless body area networks (WBANs). It addresses the specific requirements of medical, healthcare, and fitness applications by defining the PHY and MAC layers suitable for wearable and implantable devices.

IEEE 802.15.7: This task group defines the PHY and MAC layers for visible light communication (VLC). It enables communication using light-emitting diodes (LEDs) and is often used for indoor positioning, smart lighting, and other applications.

3.4.2. IEEE 802.15.4

IEEE 802.15.4 is a subgroup of features that refers to physical and medium access control layers that can support ZigBee and 6LoWPAN upper. IEEE 802.15.4 focuses on physical and data link layer specifications while ZigBee Alliance aims to provide the upper characteristics [33]. It is a standard that defines PHY and MAC layers for personal area networks that demand low rate and low cost applications. This also called a LR-WPAN protocol and has some advantages. Among them are a simple and flexible protocol stack, low cost, low energy consumption, short-range operation, reliable data transfer, and ease of operation [34]. These features are more important when operating in the Personal Operating Space (POS) also defined as Personal Area Network (PAN) that involves the human body.

Physical Layer

The 802.15.4 standard supports an extensive number of PHY options that range from

2.4 GHz to sub-GHz frequencies in ISM bands. (ISM bands are discussed earlier in this chapter.) The original IEEE 802.15.4-2003 standard specified only three PHY options based on direct sequence spread spectrum (DSSS) modulation. DSSS is a modulation technique in which a signal is intentionally spread in the frequency domain, resulting in greater bandwidth. The original physical layer transmission options were as follows:

- 2.4 GHz, 16 channels, with a data rate of 250 kbps
- 915 MHz, 10 channels, with a data rate of 40 kbps
- 868 MHz, 1 channel, with a data rate of 20 kbps

You should note that only the 2.4 GHz band operates worldwide. The 915 MHz band operates mainly in North and South America, and the 868 MHz frequencies are used in Europe, the Middle East, and Africa. IEEE 802.15.4-2006, 802.15.4-2011, and

IEEE 802.15.4-2015 introduced additional PHY communication options, including the following:

- OQPSK PHY: This is DSSS PHY, employing offset quadrature phase-shift keying (OQPSK) modulation. OQPSK is a modulation technique that uses four unique bit values that are signaled by phase changes. An offset function that is present during phase shifts allows data to be transmitted more reliably.

- BPSK PHY: This is DSSS PHY, employing binary phase-shift keying (BPSK) modulation. BPSK specifies two unique phase shifts as its data encoding scheme.

- ASK PHY: This is parallel sequence spread spectrum (PSSS) PHY, employing amplitude shift keying (ASK) and BPSK modulation. PSSS is an advanced encoding scheme that offers increased range, throughput, data rates, and signal integrity compared to DSSS. ASK uses amplitude shifts instead of phase shifts to signal different bit values.

These improvements increase the maximum data rate for both 868 MHz and 915 MHz to 100 kbps and 250 kbps, respectively. The 868 MHz support was enhanced to 3 channels, while other IEEE 802.15.4 study groups produced addendums for new frequency bands. For example, the IEEE 802.15.4c study group created the bands 314–316 MHz, 430–434 MHz, and 779–787 MHz for use in China.

Figure shows the frame for the 802.15.4 physical layer. The synchronization header for this frame is composed of the Preamble and the Start of Frame Delimiter fields. The Preamble field is a 32-bit 4-byte (for parallel construction) pattern that identifies the start of the frame and is used to synchronize the data transmission. The Start of

Frame Delimiter field informs the receiver that frame contents start immediately after this byte.

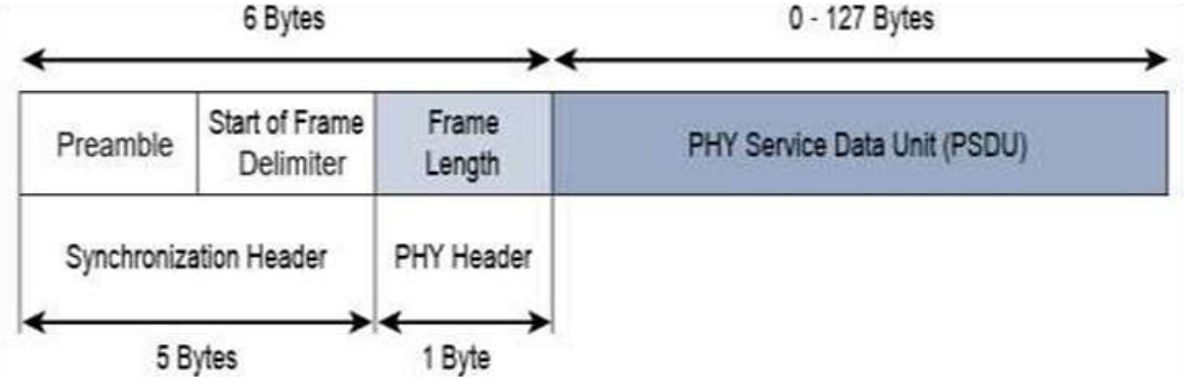


Fig.3.9: PHY header

The PHY Header portion of the PHY frame shown in Figure 4-5 is simply a frame length value. It lets the receiver know how much total data to expect in the PHY service data unit (PSDU) portion of the 802.4.15 PHY. The PSDU is the data field or payload.

MAC Layer

The IEEE 802.15.4 MAC layer manages access to the PHY channel by defining how devices in the same area will share the frequencies allocated. At this layer, the scheduling and routing of data frames are also coordinated. The 802.15.4 MAC layer performs the following tasks:

- Network beaconing for devices acting as coordinators (New devices use beacons to join an 802.15.4 network)
- PAN association and disassociation by a device
- Device security
- Reliable link communications between two peer MAC entities

The MAC layer achieves these tasks by using various predefined frame types. In fact, four types of MAC frames are specified in 802.15.4:

- Data frame: Handles all transfers of data
- Beacon frame: Used in the transmission of beacons from a PAN coordinator
- Acknowledgement frame: Confirms the successful reception of a frame
- flAC command frame: Responsible for control communication between devices

Each of these four 802.15.4 MAC frame types follows the frame format shown in Figure . In Figure, notice that the MAC frame is carried as the PHY payload. The

802.15.4 MAC frame can be broken down into the MAC Header, MAC Payload, and MAC Footer fields.

The MAC Header field is composed of the Frame Control, Sequence Number and the Addressing fields. The Frame Control field defines attributes such as frame type, address- ing modes, and other control flags. The Sequence Number field indicates the sequence identifier for the frame. The Addressing field specifies the Source and Destination PAN Identifier fields as well as the Source and Destination Address fields.

The MAC Payload field varies by individual frame type. For example, beacon frames have specific fields and payloads related to beacons, while MAC command frames have different fields present. The MAC Footer field is nothing more than a frame check sequence (FCS). An FCS is a calculation based on the data in the frame that is used by the receiving side to confirm the integrity of the data in the frame. IEEE 802.15.4 requires all devices to support a unique 64-bit extended MAC address, based on EUI-64. However, because the maximum payload is 127 bytes,

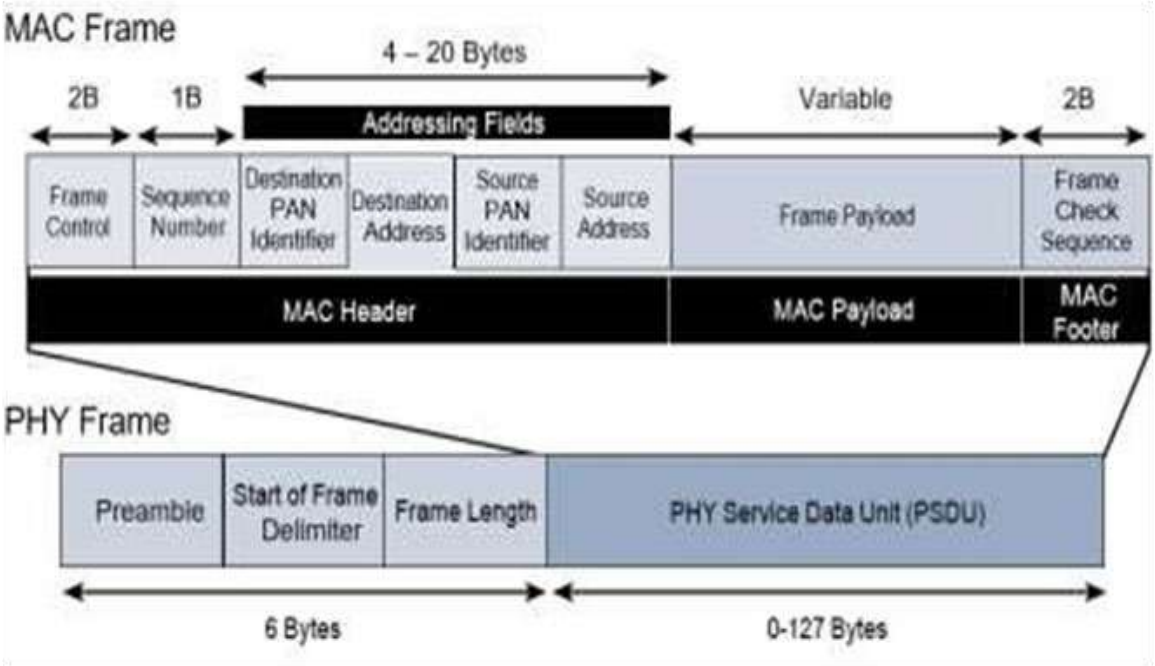


Fig.3.10: IEEE 802.15.4 MAC Format

802.15.4 also defines how a 16-bit "short address" is assigned to devices. This short address is local to the PAN and substantially reduces the frame overhead compared to a 64-bit extended MAC address. However, you should be aware that the use of this short address might be limited to specific upper-layer protocol stacks.

The IEEE 802.15.4 BE and NBE operational modes have being strongly investigated over recent years. Thus, some limitations have been addressed and the most important ones are the unbounded delay, low communication efficiency, low interference robustness, and/or fading and main powered relay nodes Figure compares IEEE 802.15.4 stack with the OSI reference model.



Fig.3.11: IEEE 802.15.4 compared with OSI reference model.

Advantages of IEEE 802.15.4:

IEEE 802.15.4 has the following advantages:

-cheap cost

-long battery life,

-Quick and simple installation

-extensible protocol stack

Disadvantages of IEEE 802.15.4:

IEEE 802.15.4's drawbacks include:

-IEEE 802.15.4 causes interference and multipath fading.

-doesn't employ a frequency-hopping approach.

-unbounded latency

-interference susceptibility

Applications of IEEE 802.15.4: IEEE

802.15.4 Applications:

-Wireless sensor networks in the industry

-Building and home automation

-Remote controllers and interacting toys

-Automotive networks

3.5. Wireless HART:

Wireless HART is a datalink protocol that operates on the top of IEEE 802.15.4 PHY and adopts Time Division Multiple Access (TDMA) in its MAC. It is a secure and reliable MAC protocol that uses advanced encryption to encrypt the messages and calculate the integrity in order to offer reliability. The architecture, as shown in fig consists of a network manager, a security manager, a gateway to connect the wireless network to the wired networks, wireless devices as field devices, access points, routers and adapters. The standard offers end-to-end, per-hop or peer-to- peer security mechanisms. End to end security mechanisms enforce security from sources to destinations while per-hop mechanisms secure it to next hop only. Wireless HART is a wireless communication standard specifically designed for industrial process automation. It is based on the Highway Addressable Remote Transducer (HART) protocol, which is widely used in industrial control systems.

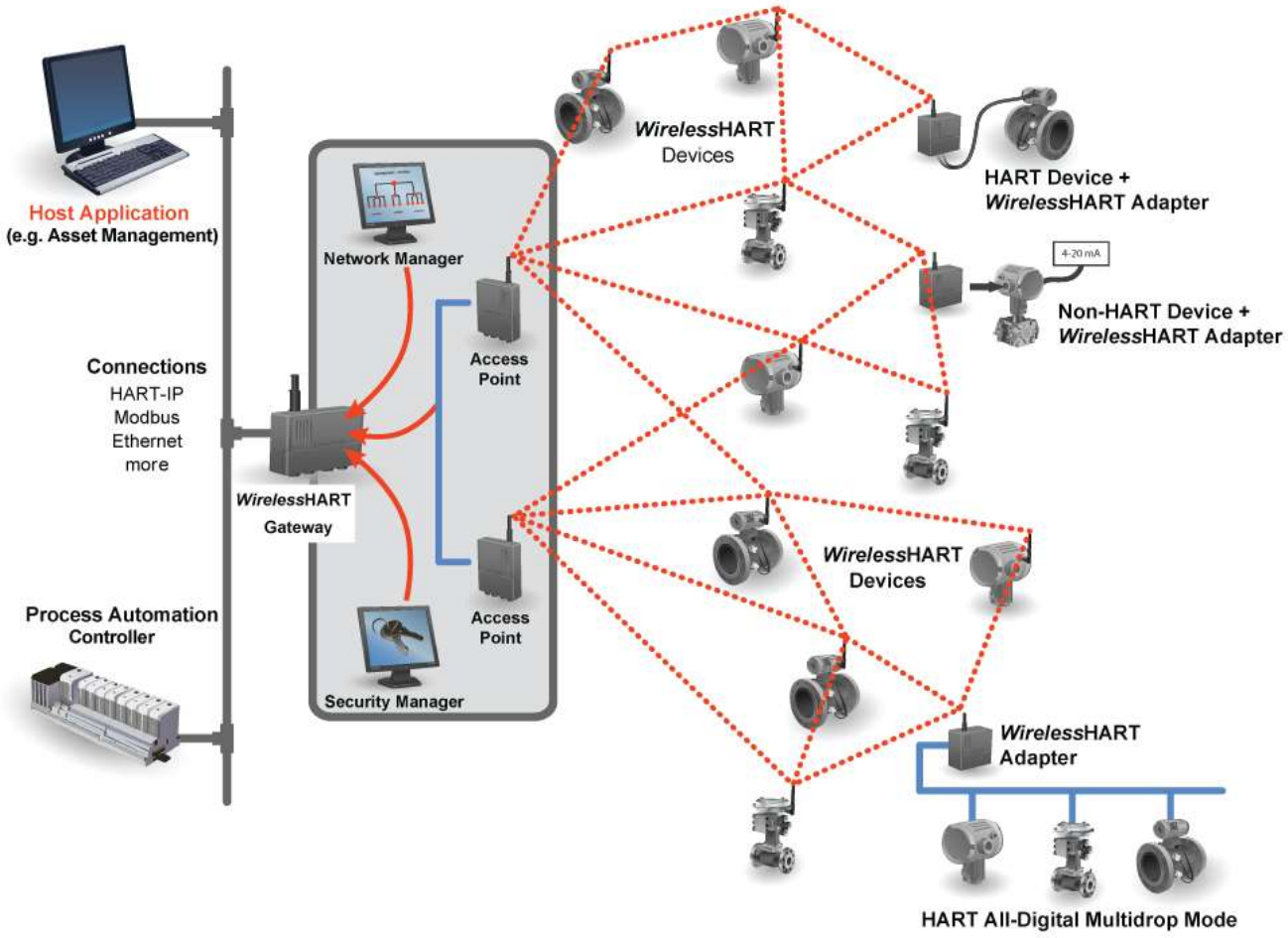


Figure.3.12: WirelessHART architecture

Wireless-HART (Highway Addressable Remote Transducer Protocol) is a variation of IEEE 802.15.4 design to work essentially as a centralized wireless network. IEEE 802.15.4 is designed to meet the requirements of industrial wireless applications with hard timing

parameter restrictions, critically security issues, and severity on obstacle interferences. The Wireless-HART protocol has the same specifications as IEEE 802.15.4 PHY, but develops its own MAC layer based on the TDMA technique. Using Bluetooth, there is no guarantee to delay values on an end-to-end wireless communication.

The absence of a hopping channel technique and a quasi-static star Bluetooth topology works against its scalability. These characteristics make them inappropriate to be used in industrial scenarios. Wireless HART comes as a solution for process control applications through the effort of some industrial organizations such as International Society of Automation 100 (ISA 100), HART, Wireless Industrial Networking Alliance (WINA) and ZigBee Alliance to attend their specific requirements ratified by the HART Communication Foundation in 2007.

Using the IEEE 802.15.4 PHY layer, Wireless-HART operates in the license-free ISM of 2.4–2.4835 GHz with 2 MHz bandwidth of each one of the 16 channels. The channels are numbered from 11 to 26 with a gap of 5 MHz between IEEE 802.11b/g adjacent channels, delivering up to 250 Kbps. Wireless-HART uses its own Time Division Multiple Access (TDMA) on the MAC layer including the
10 ms synchronized time slot features. These characteristics allow the messages routing through a network topology obstacle and interference. This is possible due to the use of self-organizing and self-healing mesh networking techniques supported by the network layer. Even being essentially a centralized wireless network, Wireless-HART uses a network manager in its stack in order to provide routing and communication schedules. This can guarantee network performance and satisfy the wireless industrial applications. The focus of Wireless-HART is communication on a one-hop level and the network layer has its responsibility to the network devices vicinity allocation.

Differing from IEEE 802.15.4, Wireless-HART uses time-synchronized the TDMA technique combined with frequency hopping on its MAC layer, thus allowing multiple devices to transmit data at the same time along different channels. During the joining process of the devices onto networks, the network manager distributes the communication links and the channel hop patterns to the devices. It also manages the enabling or disabling of the use of channels that are frequently affected by considerable interference levels, calling this feature channels blacklist,wHart-n-802-15- 4e,petersen2011wirelesshart. The eight types of devices defined on Wireless-HART are: routers, gateways, adapters, network managers, network security devices, access points and field devices on a mesh topology. All of them support the implementation of features to attend network creation, maintenance issues, data and signaling routing capability, and a minimum of reliability. A comparison between the OSI reference model and Wireless-HART protocol layers and its main features is shown in Figure 3.13.

OSI Layer	HART
Application	Command Oriented. Predefined Data Types and Application Procedures
Transport	Auto-Segmented transfer of large data sets, reliable stream transport, Negotiated Segment sizes
Network	Power-Optimized Redundant Path, Mesh to the edge Network
Data Link	Secure, Time Synched TDMA/CSMA, Frequency Agile with ARQ
Physical	2.4 GHz Wireless 802.15.4 based radios, 10dBm Tx Power

Figure3.13: Protocol stack of Wireless HART

Another addressable characteristic of Wireless-HART is the information blocks that each network device maintains on its memory. The information of neighbor nodes and the next reachable device is called a neighbor information block. The connection with the network layer is made through the block information, adding data to the network layer routing table. Working with TDMA as a medium access technique, the network devices have very stringent timing requirements to accomplish network synchronization premises. This happens because synchronization occurs both in the joining process and in normal operations.

Wireless HART technology allows users to access the vast amount of unused information stranded in these installed HART smart devices—85% of the installed HART devices. It also provides a simple, reliable and secure way to deploy new points of measurement and control without the wiring costs.

1. Simple:

Wireless HART is a robust technology that is simple to implement. It enables users to quickly and easily gain the benefits of wireless technology while maintaining compatibility with existing HART devices, tools and systems.

- Easy Installation and Commissioning:-

-Familiar tools, work flow and procedures

-Multiple power options

-Reduced installation and wiring costs

-Coexistence with other wireless networks

-Supports both star and mesh topologies

-Add devices one at a time

- Automatic Network Features:-

-Self-organizing and self-healing

-Always-on security

-Adjusts as new instruments are added

-Adjusts to changes in plant infrastructure

2. Reliable:

Industrial facilities with dense infrastructures, frequent movement of large equipment, changing conditions, or numerous sources of radio-frequency and electromagnetic interference may have communication challenges. Wireless HART includes several features to provide built-in 99.9% end-to-end reliability in all industrial environments.

- Standard Radio with Channel Hopping

-Radios comply with IEEE 802.15.4-2006

-2.4GHz license free frequency band

-"Hops" across channels to avoid interference

-Delivers high reliability in challenging radio environments

- Coexistence with Other Wireless Networks

-Clear Channel Assessments tests for available channels

-Blacklisting avoids frequently used channels

-Optimizes bandwidth and radio time

-Time synchronization for on-time messaging

- Self-Healing Network

-Adjusts communication paths for optimal performance

-Monitors paths for degradation and repairs itself

-Finds alternate paths around obstructions

-Mesh network and multiple access points

3. Secure: Wireless HART employs robust security measures to protect the network and secure the data at all times. These measures include the latest security techniques to provide the highest levels of protection available.

- Protects Valuable Information

-Robust, multi-tiered, always-on security

-Industry standard 128-bit AES encryption

-Unique encryption key for each message

-Data integrity and device authentication

-Rotate encryption keys used to join the network

- Protects Wireless Network

-Channel hopping

-Adjustable transmit power levels

-Multiple levels of security keys for access

-Indication of failed access attempts

-Reports message integrity failures

-Reports authentication failures

-Safe from Wi-Fi type Internet attacks

3.6. Z-Wave:

Z-Wave is a low-power MAC protocol designed for home automation and has been used for IoT communication, especially for smart home and small commercial domains. It covers about 30-meter point- to-point communication and is suitable for small messages in IoT applications, like light control, energy control, wearable healthcare control and others. It uses CSMA/CA for collision detection and ACK messages for reliable transmission. It follows a master/slave architecture in which the master control the slaves, send them commands, and

handling scheduling of the whole network. Z-Wave was developed and is overseen by the company Zensys to provide wireless communication between devices with a focus on residential automation. Monitoring and controlling of lighting, ambient temperature and security through sensors and actuators by tablets, smartphones or computers are some applications in its portfolio. Z-Wave devices are arranged in mesh networktopology. They can send and receive messages from any device that is connected to the network.

The protocol is a proprietary standard based on the ITU G.9959 specification that operates in the Industrial, Scientific, and Medical (ISM) radio frequency band. Z-Wave transmits on 868.42 MHz (Europe) and 908.42 MHz (United States) frequencies working with FSK and Gaussian Fase Shift Keying (GFSK) modulations. With low transmission rates of 9.6 Kbps, 40 Kbps and 100 Kbps, it employs symmetric AES-128 encryption. The MAC layer uses the CSMA-CA technique for a medium access control technique and, based on ITU G.9959, has the following characteristics: a capacity of 232 unique network identifiers that allows the same quantity of nodes joining the network; collision avoidance mechanism; back-off time when collision occurs; reliability guaranteed by receiving acknowledgments; frame validation and retransmission mechanisms. A power saving mechanism is achieved due to a sleep mode with a dedicated wake-up pattern Figure depicts the Z-Wave protocol stack.

The Z-Wave basic device classes are the following: Portable Controller, Static Controller, Slave, and Slave with Routing Capabilities. Different classes provide the device with a certain role in the Z- Wave network. Inside a Basic Class, Generic and Specific device classes are used to achieve the wanted functionality in the control network. In the Z-Wave protocol, the unique identification of the devices is used through a 32-bit ID. This ID value cannot be changed as it is written in the device chipset by the device manufacturer. A Z-Wave network has only one primary controller device at a time.

Each of the 232 nodes of this network can also be a repeater for forwarding data to its neighbors, mediating a connection. Battery-powered nodes do not enjoy this facility. In an environment with a certain level of device drift or even when a device is removed from the network for some reason, the network topology may change. Changing network topology can lead to problems in packet forwarding and packet routing in the network. To minimize this effect, routing tables should be kept up-to-date, optimized and any new topology detected; Z-Wave supports the discovery and suitability of the new network topology.

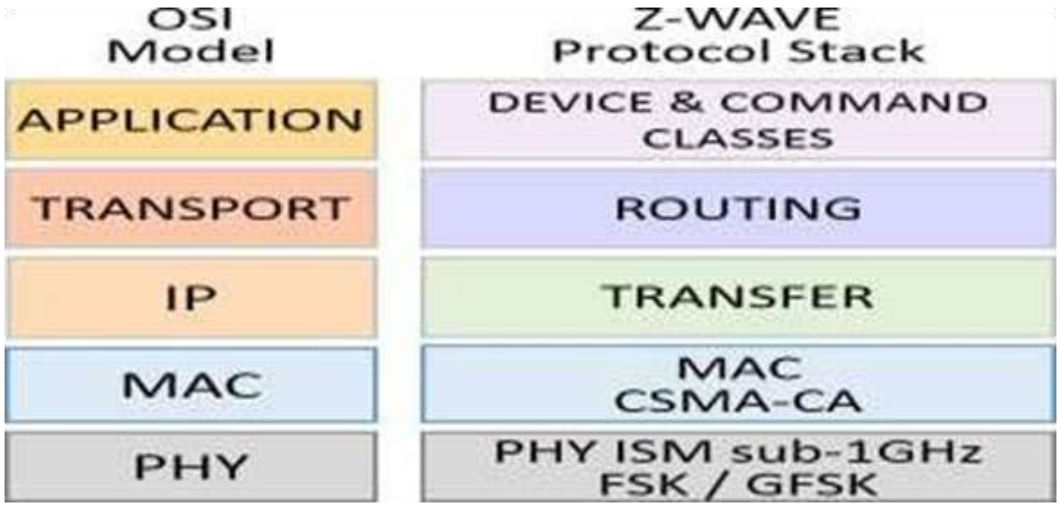


Figure 3.14: Z-Wave protocol stack.

This is possible by keeping the routing table up-to-date on each device and showing all neighboring devices. When a node changes its position or is removed from the network, a topology failure can start an automatic topology and healing procedure to detect the new topology and define the best routes to update the routing tables. This mechanism is subjected to unauthorized modification of routing table attacks by rouge nodes.

The transfer (or transport) layer management functions are: communication between two neighbor nodes, packet acknowledgment, low power network nodes awake (Beaming), and packet origin authentication. This layer controls the Beam frames used to wake-up battery powered Z-Wave devices, as each primary controller device of a cluster can handle up to 232 nodes. All nodes can act as a packet repeater, except those devices that are batteries powered. This is Z-Wave mesh topology formed. Z-Wave data security is based on AES and on the cipher block chaining message authentication code (CBC-MAC). However, standards and rules for command classes, device types and timers are missing. These characteristics are only acquired in the new advanced security framework (S2) determined by the Z-Wave Alliance and developed in conjunction with the cyber security community. For the certification of new products as of 2017, Z-Wave brings devices a higher level of security. The structure of S2 is based on the protection of the devices that is already associated with the network, so they are not hacked while still connected to the network. Once the device has already been associated to the network through its pin-code or QR (Quick Response) code, there is an exchange of security keys through the Elliptic Curve Diffie- Hellman (ECDH) algorithm.

3.7. DASH7:

DASH7 is a wireless communication protocol for active RFID that operates in globally available Industrial Scientific Medical (ISM) band and is suitable for IoT requirements. It is mainly designed for scalable, long range outdoor coverage with higher data rate compared to traditional ZigBee. It is a low-cost solution that supports encryption and IPv6 addressing. It supports master/slave architecture and is designed for burst, lightweight, asynchronous and transitive traffic. Its MAC layer features can be summarized as follows

Filtering: Incoming frames are filtered using three processes; cyclic redundancy check (CRC) validation, a 4-bit subnet mask, and link quality assessment. Only the frames that pass all three checks are processed further.

Addressing: DASH7 uses two types of addresses: the unique identifier which is the EUI-64 ID and dynamic network identifier which is 16-bit address specified by the network administrator.

Frame format: The MAC frame has a variable length of maximum 255 bytes including addressing, subnets, estimated power of the transmission and some other optional fields.

The RFID technologies such as NFC and Dash7 are widely used in WSN (wireless sensor networking). Most of the Smartphone will have these technologies integrated to provide many facilities to the users. These include building access, mobile payments, advanced location services, home automation, ticketing and more. This means that Dash7 will be part of IoT(Internet of Things) which acquires sensors data and use that to manage social network applications.

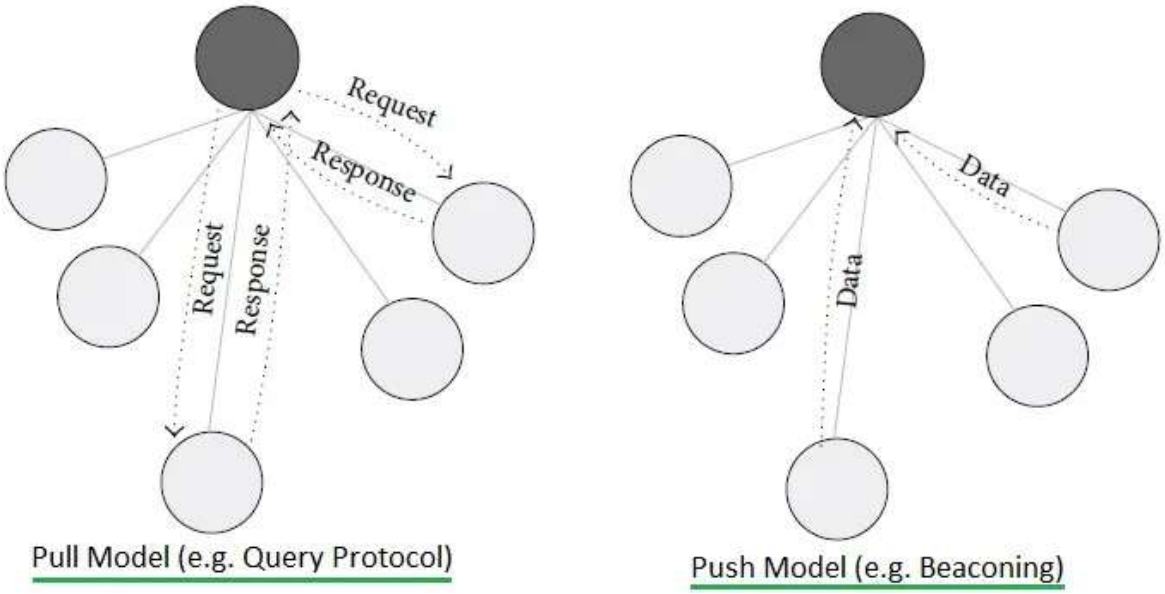


Figure.3.15: DASH7 Communication Models

There are four different device classes defined in D7A (Dash7 Alliance Protocol).

Blinker: It only transmits and does not use a receiver.

Endpoint: It can transmit and receive the data. It also supports wake-up events.

Sub controller: It is full featured device. It is not always active. It uses wake on scan cycles similar to end points.

Gateway: It connects D7A network with the other network. It will always be online. It always listens unless it is transmitting.

D7A describes full functional RFID tag. All the devices in Dash7 network support one or more of the above-mentioned device classes.

DASH7 supports two communication models: pull and push.

The dialogs between tags and interrogators are query response based (referred as pull model). This request response mechanism is described by the D7A Query Protocol Data transfer initiated from the tags to the gateway on the other hand is based on the push model. Both of these models are depicted in the figure. This approach for instance is implemented as an automated message or beacon which is sent on specific time intervals. This system is called Beacon Transmit Series.

DASH7 defines two types of frames viz. a foreground frame and a background frame. The foreground frames are regular messages which contain data or data requests. Background frames on the other hand are very short broadcast messages. Background frames are used by the D7A Advertising Protocol for rapid ad-hoc group synchronization.

Background frame = { subnet(1 byte), payload(3 bytes), CRC16(2 byte) } Foreground frame = {length(1 byte), headers(3-38 bytes), payload(0 to 249 bytes), Footer(0 to 20 bytes), CRC16(2 bytes) }

DASH7 (Data and Sensing Hierarchy 7) is an open standard and wireless communication protocol designed for low-power, long-range, and low-data-rate applications in the Internet of Things (IoT) domain. It focuses on providing reliable communication with extended range and efficient power consumption for battery-operated devices.

3.8. Zigbee Smart Energy:

ZigBee Smart Energy (SE) is a standard for interconnecting and interoperating devices, via radio frequency, directed towards monitoring, managing and automating energy, gas and water usage. It seeks to be a useful tool for creating "Green Homes", and is aimed at coordinating energy usage, optimizing its generation and consumption.

ZigBee SE is a world-leading standard widely used for smart metering (electricity, gas and water) and home automation (wireless domotics). ZigBee is a simple data transmission protocol designed to be used as a low rate wireless personal area network (LR-WPAN). Based on the IEEE 802.15. ZigBee smart energy is designed for a large range of IoT applications including smart homes, remote controls and healthcare systems. It supports a wide range of network topologies including star, peer-to-peer, or cluster-tree.

A coordinator controls the network and is the central node in a star topology, the root in a tree or cluster topology and may be located anywhere in peer-to-peer. ZigBee standard defines two stack profiles: ZigBee and ZigBee Pro. These stack profiles support full mesh networking and work with different applications allowing implementations with low memory and processing power. ZigBee Pro offers more features including security using symmetric-key exchange, scalability using stochastic address assignment, and better performance using efficient many-to-one routing mechanisms.

Wireless protocol for device monitoring and control: Zigbee Smart Energy (Zigbee SE) is a protocol designed for monitoring and actively managing energy consumption at the end-user level. For both utilities and consumers, Zigbee SE can help reduce waste, energy consumption and enables utilities to monitor and manage customers' energy use. Furthermore, the end-user can monitor their energy consumption.

Affordable and easy to use

With Zigbee SE it easier to monitor energy consumption and lower the energy costs – for both utilities and the end-user. Zigbee SE is easy to use because it is interoperable with other Zigbee protocols and works across manufacturers. The standardized protocol provides fast access to new markets for solution providers, and it reduces research costs

Reducing energy consumption

Intelligent energy management with Zigbee SE also benefits the environment, because it improves energy availability. With the access to monitor the energy consumption, both utilities and end-users can regulate the use of energy, thus saving money and reducing the impact on the environment. Increased energy efficiency reduces the need for additional generation plants.

Zigbee Smart Energy features

The Zigbee SE offers real-time information, tracking of energy consumption, multiple control methods e.g. emergency signals and duty cycling, and includes the ability to randomize start and end times to avoid energy spikes. The protocol naturally uses data encryption to safeguard data. The protocol fits multiple commodities, e.g. electricity, gas, and water, and it has several measurement types such as load profile, power factor, demand, and tiers.

ZigBee SE is a world-leading standard widely used for smart metering (electricity, gas and water) and home automation (wireless domotics). ZigBee is a simple data transmission protocol designed to be used as a low rate wireless personal area network (LR-WPAN). Based on the IEEE 802.15.4 specification, for a set of high-level communications protocols, it's a low-powered, low-bandwidth digital radio communication system. Among its most important applications are automation in the home and smart metering. The protocol is intended for devices requiring low-power, relatively low transmission speeds and short distances between devices. In terms of transmission speeds, in the best-case scenario and depending on the frequency used, it can reach 250 Kb/s. The maximum transmission distance can vary from 10 to 100 meters, depending on the frequency, transmission power and environmental conditions. Radio channel access is provided by CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) – a low level network access control protocol that permits multiple stations to use the same transmission medium. ZigBee devices consume very small amounts of energy and are very low cost. They use protocols from different layers (PHY, MAC, network, security, and application). The network, security and application layers are defined by ZigBee Alliance. The networks can work on different frequencies: 868 MHz, 915 MHz and 2.4 GHz. ZigBee networks support around 65,000 devices.

ZigBee networks support star and mesh topologies. Three types of devices are defined:

1) **Coordinator:** this is the device that coordinates and forms the network, which means that every network must always have one. Once this device creates the network, other devices (Routers or End devices) can join. It is responsible for selecting the frequency channels and assigning network identifiers (PAN ID) to devices. The PAN ID is used to communicate between network devices. The coordinator can help to route data over mesh networks. It requires a permanent power supply, must always be active and be able to support child devices.
2) **Router:** First, it must join the network, after which it can allow other Routers and End devices to join. It requires a permanent power supply, must always be active and be able to support child devices.
3) **End Devices:** These do not connect to other network devices. They are usually battery-powered devices and can go into "sleep" mode to save energy.

The following image is a graphical representation of a mesh network structure:

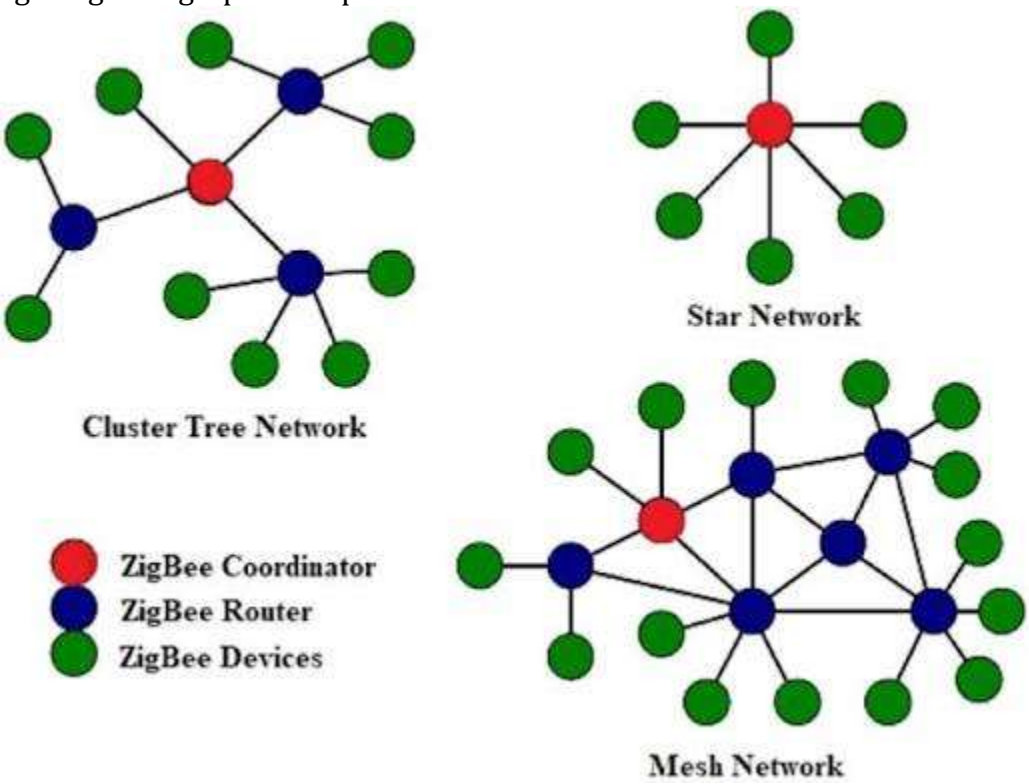


Figure.3.17: Zigbee smart energy architecture

A number of standards use ZigBee as a base. The most common of these are:

ZigBee Home Automation (ZHA): This is a home automation-oriented global standard for controlling applications like lighting, temperature control, energy management, security and accident prevention. ZigBee Smart Energy (ZSE): This is a global standard that allows service providers and Home Area Network (HAN) electricity distribution companies to manage energy consumption. ZSE also allows suppliers and customers to interact, so that both can access smart communications. ZigBee Light Link (ZLL): This is a global standard that permits consumer lighting elements and other elements to

interoperate with each other, giving consumers wireless access to these elements. It allows consumers to control their home lighting, while managing energy use and making their homes “greener”.

Reason for ZigBee Smart Energy

ZigBee SE provides service providers and power distributors with a simple wireless access network within homes (Home Area Network, or HAN). Smart Energy offers these groups and their customers the possibility of communicating with each other directly in order to control smart applications (e.g., thermostats and other devices used to control high energy use in the home). Having access to customers’ instantaneous consumption enables power distributors to more efficiently manage the electricity smart grid (generation and distribution). Furthermore, customers can receive real-time information on their energy use through devices installed inside the home, as well as by accessing the HAN through the services provided by energy distributors and/or service providers.

3.9. Bluetooth Low Energy:

Bluetooth low energy or Bluetooth smart is a short range communication protocol with PHY and MAC layer widely used for in-vehicle networking. Its low energy can reach ten times less than the classic Bluetooth while its latency can reach 15 times. Its access control uses a contention-less MAC with low latency and fast transmission. It follows master/slave architecture and offers two types of frames: adverting and data frames. The Advertising frame is used for discovery and is sent by slaves on one or more of dedicated advertisement channels. Master nodes sense advertisement channels to find slaves and connect them. After connection, the master tells the slave it’s waking cycle and scheduling sequence. Nodes are usually awake only when they are communicating and they go to sleep otherwise to save their power.

Being part of the Bluetooth v4.0 standard adopted in 2010-06-30, Bluetooth Low Energy (BLE) is also known as Smart Bluetooth. BLE is an IEEE 802.15.1 variation with better and more suitable capacities for low power applications than the classic Bluetooth Basic Rate. Devices that demand communication with both standards of Bluetooth are required to implement and support both protocol stacks due the incompatibilities among them. Star is the only topology accepted by BLE due the standard definition that does not permit physical link connections among slave devices. Any data exchanged between two slave devices shall pass through the unique master and a slave device may not be

connected to two master units at the same time. These premises define the formation of a BLE star pico-net.

Using a similar protocol stack as classic Bluetooth, the differences between them starts above the L2CAP layer. Above the L2CAP layer, BLE is the application layer that uses a set of functionalities, which are not present in the classic Bluetooth specifications. These functionalities are the Attribute Protocol (ATT), the Generic Attribute Profile (GATT), the Security Manager Protocol (SMP) and the Generic Access Profile (GAP). Figure 7 depicts the BLE protocol stack.

The two main roles of BLE are: controller and host. BLE differs from the classical Bluetooth in the controller stack that defines the association methods of the devices. A slave can belong to only one pico-net during an association lifetime, and is synchronized with only one master element.

A Host Controller Interface (HCI) is a communication standard applied between the slave and controller. In the Bluetooth Basic Rate, 79 channels are used with a 1 MHz bandwidth to reduce interference with adjacent channels. In Bluetooth Low Energy, the channels are defined in the 2.400– 2.4835 GHz band with a 2 MHz guard band. To achieve scalability, the master device controls the number of hosts associated with it by adjusting the value of the connection interval (ConnInterval parameter) between hosts and controllers.

Link layer manages events generated by the hosts, at determined time intervals, using the advertising channels. Bidirectional data flow is obtained with a connection between elements, when slaves advertising packets are received by master elements. The energy save handling done at MAC layer can put the slaves in a sleeping mode by default and waking them periodically through a Time Division Multiple Access (TDMA) scheme.

In the classic Bluetooth basic protocol, this layer a stop-and-wait flow control mechanism is used to provide error recovery capabilities. At BLE, the L2CAP is an adaption of the classic Bluetooth basic protocol stack but optimized and simplified to receive the application layers designed for low energy platforms. Data exchange between the application layer and link layer are also done by L2CAP using no retransmission techniques or flow control mechanisms as used on the classic Bluetooth. Not using retransmission or flow control mechanisms (present in the classic Bluetooth) and segmentation and reassembly capabilities, the Packet Data Units (PDU) (limited to 23 bytes in BLE) received by the application layer is delivered ready to fit the maximum size of the L2CAP payload.

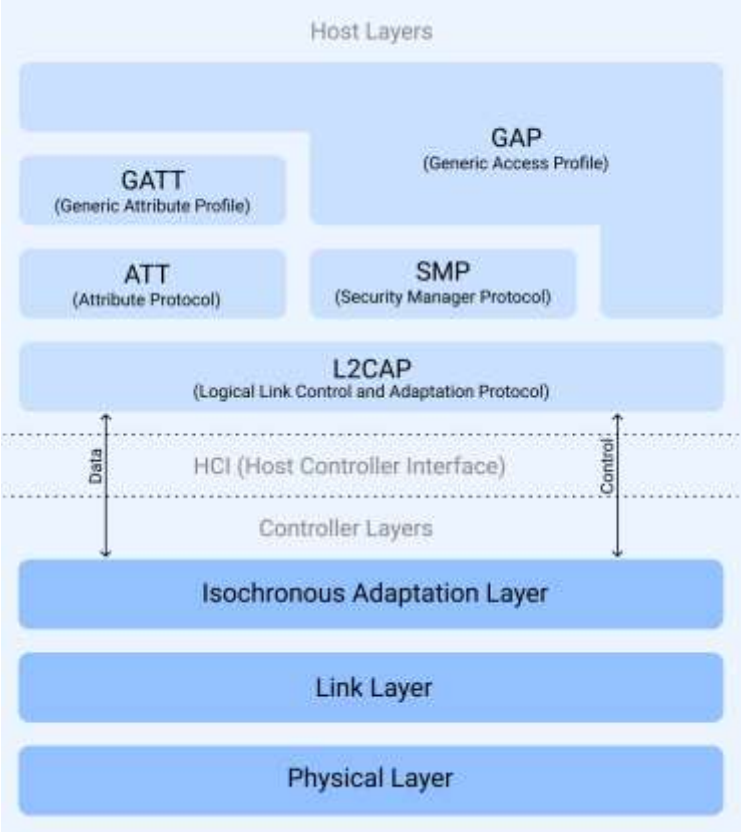


Figure.3.19: Bluetooth low energy protocol stack.

When two devices are connected under a server and client association architecture, the server needs to maintain a set of attributes. The Attribute Protocol (ATT) handles the attributes of this connection like the definition of data structure used to store the information managed by the Generic Attribute Profile (GATT) that works on top of the ATT. GATT defines the client or server functionalities of a connection and this association is independent of the master or slave roles. The attributes of the server need to be accessed by the client through the requests sent, which trigger the response messages of the server. It is also possible for a server to send to a client, unsolicited messages like notifications that do not need any confirmation message to be sent by the client. A server is also required to send indication messages, which need

confirmation messages to be sent by the client. The slave sends requests for responses and indications prior to transactions confirmation following a stop-and-waitscheme. Slaves can either write attributes values at the master.

A framework defined by GATT performs the role of discovery services using the ATT attributes, and allows exchange of characteristics between devices interconnected. An attribute carries a set of characteristics that includes a value and properties of the parameter monitored by the device. For example, a humidity sensor needs humidity characteristics and attributes to describe this sensor, and to store its measurements. Thus, this sensor needs a further attribute to specify the measurement units. Creating specific profiles with the Low Energy Bluetooth standard takes place in the Generic Attribute

Profile (GATT). GATT uses the Attribute Protocol (ATT) protocol in addition to the lower stack protocols, in order to introduce the subdivision of retained server attributes into services and features. Services can contain a set of features, which can include a single value (accessible from the client) and other numerical data that describe such features. Among the assignments of GAP profile specifications are: device role rights, discovery devices and services, as well as establishing connections and security. A new profile based on the existing profile requirements can be created following a profile hierarchy. The interoperability of different devices can be handled through application profiles.Bluetooth is designed to offer a low-cost alternative to Wi-Fi at the expense of the transmission range. Its transmission range is considerably shorter (up to 100 m LOS) and data rate does not exceed 721.2 Kbps in the classic Bluetooth Basic Rate version and can reach 3 Mbps with the Enhanced DataRate feature. BLE operates at 1Mbps rate on its physical layer, while its application layer can handleonly 236.7 Kbps.

In Bluetooth Low Energy, there are no subdivisions in power classes but only the maximum and minimum output power values of the transmitter are provided. Only an approximate value of the maximum reachable distance can be predicted. The low power required for transmission is the main feature of the Bluetooth Low Energy standard and this result is due to enhancements made on the classic version. These enhancements include reduced frequency band and shorter PDU packets. An energy evaluation is offered at using CC2640 radio chipset consumption reference measurements. The comparison is made when operating on 0 dBm transmission power by gathering the main characteristics of Bluetooth and BLE.

Bluetooth v5.0 has no functional block included in its first and second layers when compared to versions v4.0, v4.1, and v4.2. A representation of the inter-layer communication structure and the relationship with Bluetooth layers of different Bluetooth versions can be seen in Figure 8. Device-to- device file transfers, wireless speakers, wireless headsets, and Body Sensor Networks are often enabled with Bluetooth versions.

Finally, some characteristics of the Bluetooth BR and BLE technologies are summarized in Figure 9. This figure allows the comparison of their differences according to the PHY and MAC layer characteristics.

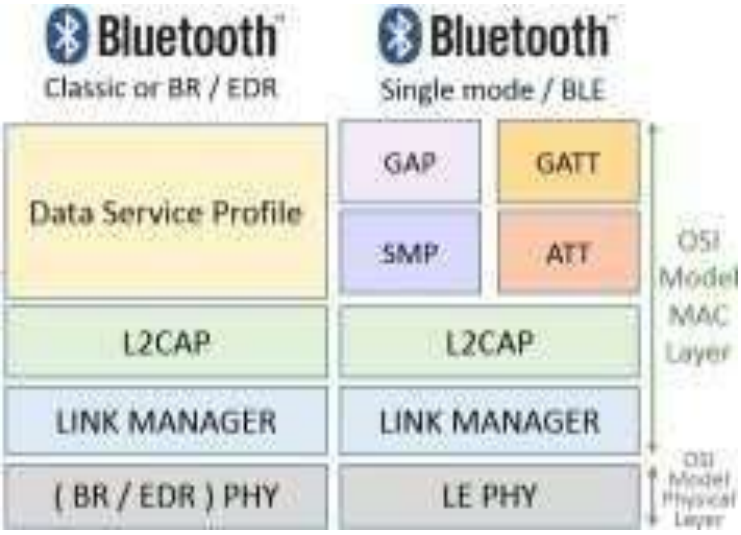


Fig: Bluetooth power class classification.

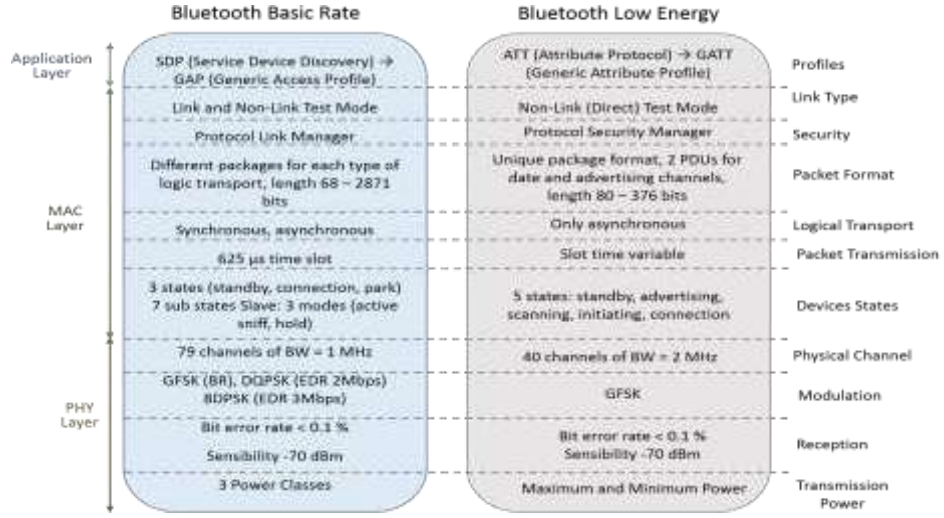


Figure.3.20: Bluetooth basic rate versus Bluetooth low energy.